Hardware Components Assignment

Day 5: Assemble and Disassemble a PC

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Task 4: Safety Precautions & Tools Checklist

Before beginning the assembly or disassembly process, it is critical to prepare the workspace and ensure all safety measures are in place.

Tools Needed:

- **Phillips Head Screwdriver (Magnetic tip preferred):** Essential for removing case screws, motherboard standoffs, and drive mounts without dropping screws into the case.
- **Anti-Static Wrist Strap or Mat:** Prevents Electrostatic Discharge (ESD) from your body, which can instantly kill sensitive microcomponents like the CPU or RAM.
- **Small Parts Tray / Cups:** Keeps different types of screws organized so they aren't lost or mixed up during reassembly.

Safety Precautions:

- **Unplug All Power:** Always disconnect the power cord from the wall and switch off the PSU to prevent electrical shock or short circuits.
- **Discharge Residual Power:** Press the power button for 5 seconds after unplugging to drain any leftover power in the motherboard capacitors.
- **Never Force Components:** If a part (like RAM or a GPU) won't come out or go in, do not force it. Check for hidden latches, clips, or screws to prevent physical damage.

Task 1: Main Components Found Inside the PC Cabinet

Upon opening the desktop cabinet, the following primary components are visible:

- **Motherboard:** The large main circuit board that connects all other components together, allowing them to communicate.
- **CPU (Central Processing Unit):** The brain of the PC, usually hidden beneath a large cooling fan or heatsink, responsible for executing all instructions.
- **RAM (Random Access Memory):** Long, thin sticks slotted into the motherboard that act as the system's short-term workspace for active applications.
- **GPU (Graphics Processing Unit / Video Card):** A bulky card plugged into a PCIe slot, responsible for rendering images, video, and 3D graphics.
- **Storage (HDD / SSD):** Drives mounted in bays or directly on the motherboard (M.2) that permanently store the operating system, games, and files.
- **PSU (Power Supply Unit):** A metal box (usually at the top or bottom rear) with a thick bundle of cables that converts wall AC power to DC power for the PC.
- **Cooling System (Fans/Heatsinks):** Case fans and CPU coolers designed to pull cold air in and push hot air out to prevent thermal throttling.

Task 2: PC Disassembly Process Documentation

Below is the step-by-step documentation of the disassembly process:

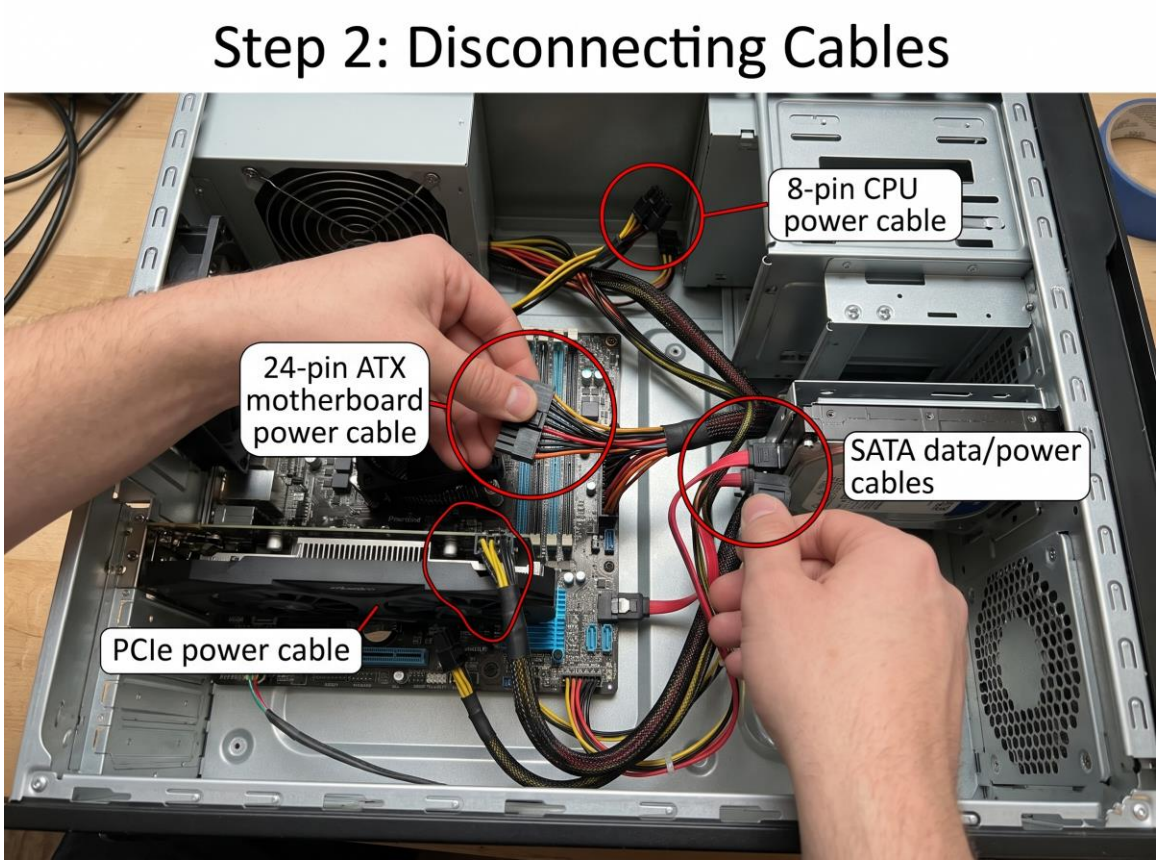
Step 1: Unplugging Power & Removing Side Panel

Disconnected the main power cable. Unscrewed the thumbscrews on the back of the case and slid the side panel off to expose the internal components.



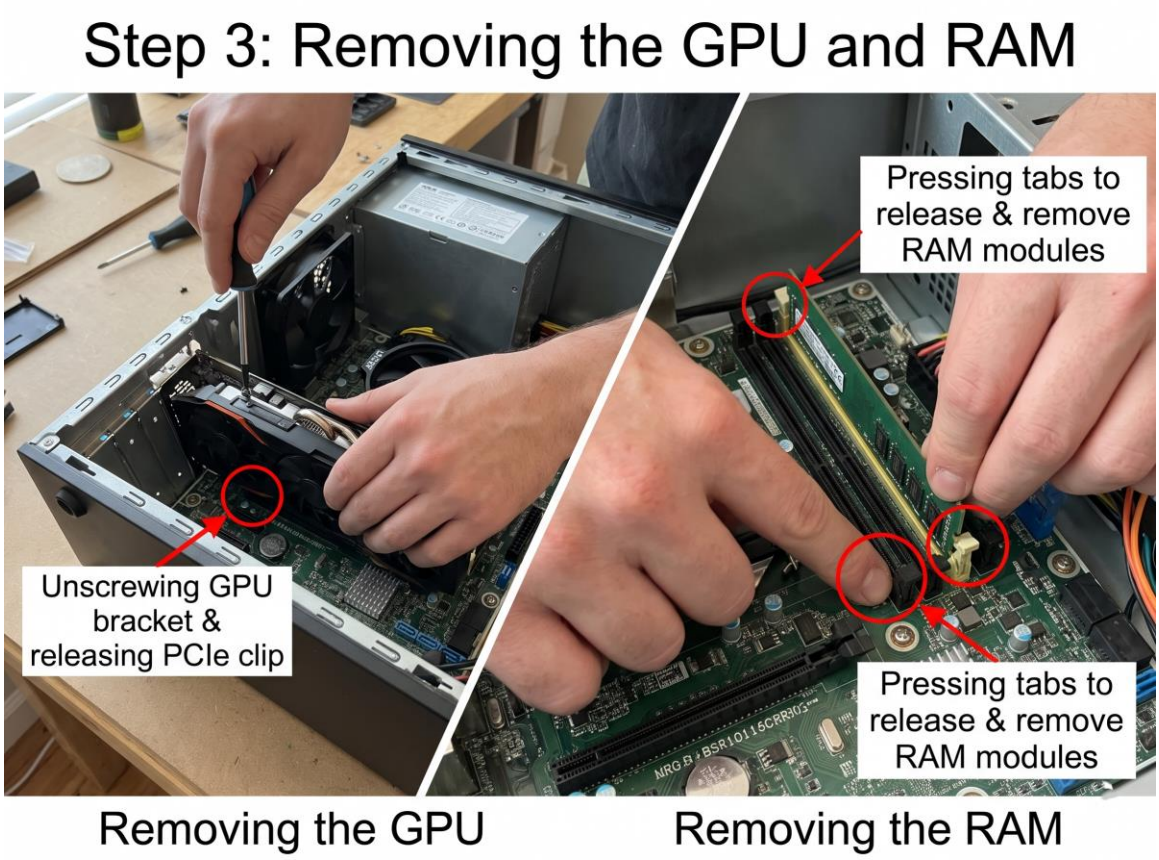
Step 2: Disconnecting Cables

Carefully unplugged the 24-pin ATX motherboard power, the 8-pin CPU power, PCIe power from the GPU, and SATA data/power cables from the storage drives.



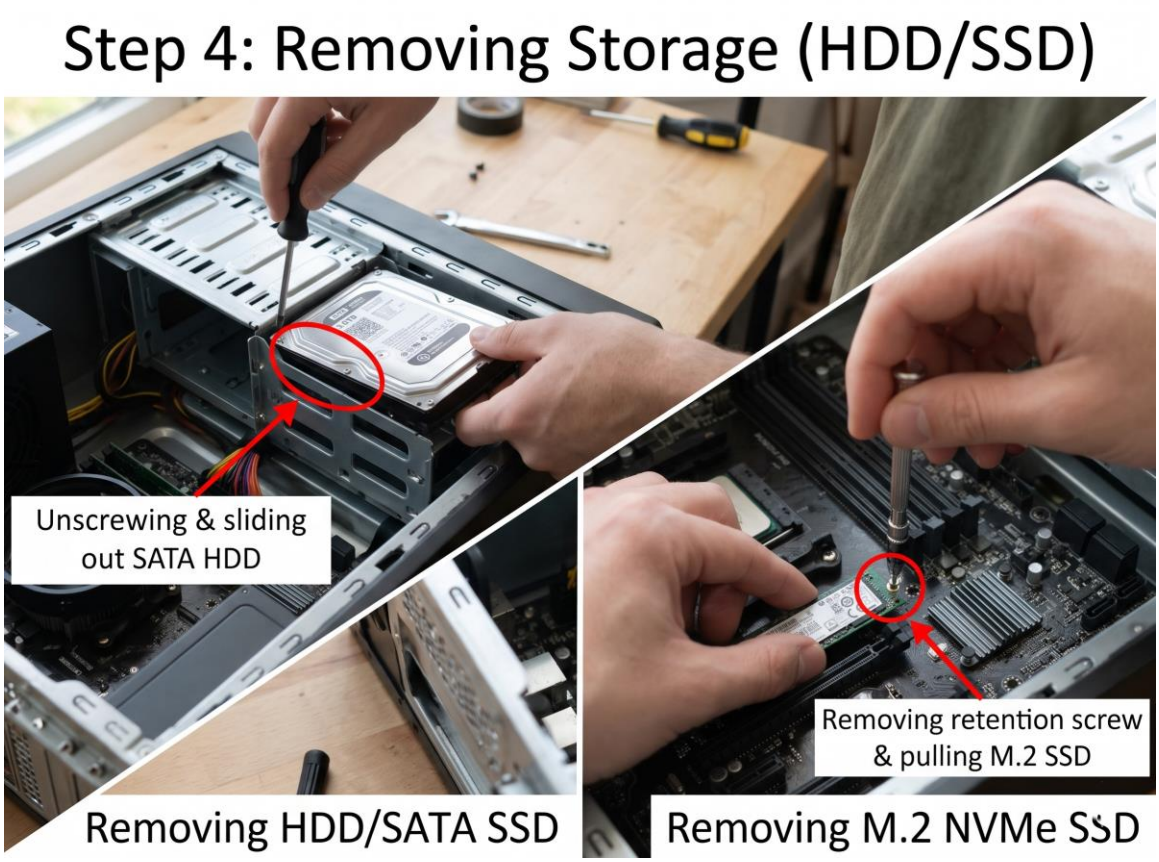
Step 3: Removing the GPU and RAM

Unscrewed the GPU bracket, pushed the PCIe retention clip down, and pulled the GPU out. Next, pressed the tabs on the RAM slots and removed the memory sticks.



Step 4: Removing Storage (HDD/SSD)

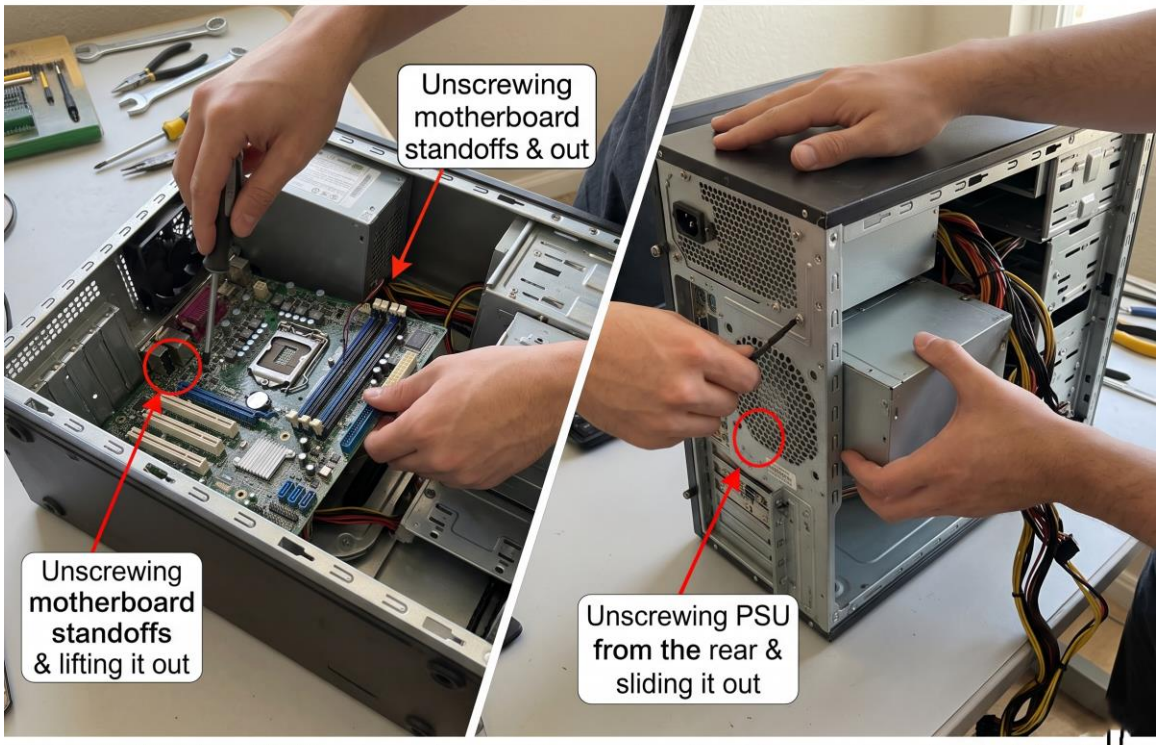
Unscrewed the HDD/SATA SSD from its drive cage and slid it out. (For NVMe, removed the tiny retention screw and pulled the drive from the M.2 slot).



Step 5: Removing Motherboard and PSU

Unscrewed the motherboard standoffs and carefully lifted the board out by its edges. Finally, unscrewed the Power Supply Unit from the rear of the case and pulled it out along with its cables.

Step 5: Removing Motherboard and PSU



Removing the Motherboard

Removing the Power Supply Unit (PSU)

Task 3: PC Assembly Process Documentation

Below is the step-by-step documentation of the reassembly process, emphasizing why this specific order makes the build easier:

Step 1: Install PSU into the Case

Connected the Power Supply Unit first. Why: It provides a solid foundation, and you can route the heavy cables out of the way before the case gets crowded.

Step 2: Core Motherboard Prep (Outside Case)

Installed the CPU, RAM, and CPU cooler onto the motherboard while it rested on a flat, anti-static surface (like its cardboard box). Why: It is much harder to apply thermal paste and push down RAM sticks when working inside the cramped case.

Step 3: Mount Motherboard into the Case

Carefully lowered the prepped motherboard into the chassis, aligning the rear ports with the I/O shield, and screwed it into the standoffs. Why: It is the central hub that everything else plugs into.

Step 4: Install Storage Drives

Mounted the HDDs/SSDs into their drive cages and secured them. Why: Easier to do before bulky components block access to the drive bays.

Step 5: Connect All Power and Data Cables

Plugged in the 24-pin motherboard, 8-pin CPU, SATA data, and front-panel connectors (power button, USB). Why: Doing cable management before installing the massive GPU ensures you have plenty of room for your hands.

Step 6: Install the GPU

Slotted the Graphics Card into the top PCIe slot, screwed it to the chassis bracket, and connected its PCIe power cables. Why: The GPU is the largest component; installing it last ensures it doesn't block your access to tiny motherboard headers.